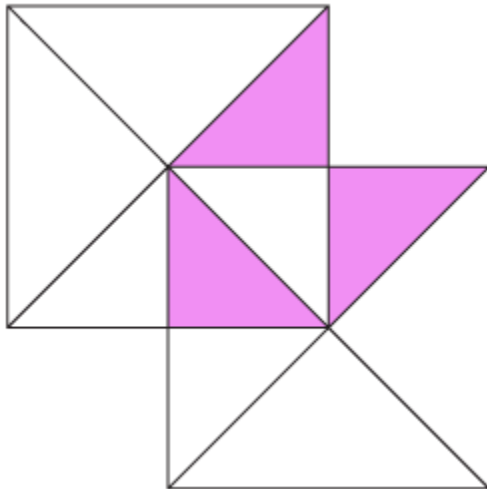


Q1.

What is the **minimum** value of the missing digit in 79____482 so that the number when rounded off to the nearest hundred thousand and ten thousand gives the same value?

Q2.

What fraction does the shaded part of the figure represent?



$$\frac{2}{16}$$

$$\frac{3}{16}$$

$$\frac{3}{8}$$

$$\frac{3}{14}$$

Q3.

Fractions p and q lie between 0 and 1 on the number line.
What can you say about the product of p and q ?

It will **always** be greater than 1.

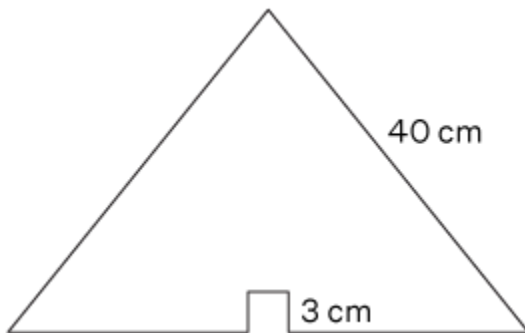
It will **always** be greater than both p and q .

It will **always** be greater than p and less than q .

It will **always** be less than both p and q .

Q4.

The side length of an equilateral triangle is 40 cm. From one of its sides, a small square of side 3 cm is removed.



What is the perimeter of the new shape formed?

- 108 cm
- 117 cm
- 126 cm
- 132 cm

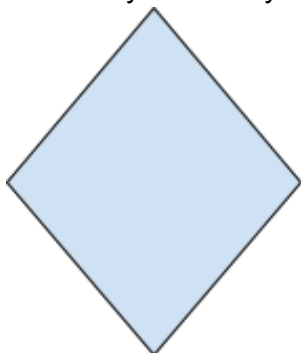
Q5.

Alex divides a number X by p . He gets 6 as the quotient and 7 as the remainder. Which of the following could be the values of X and p ?

- $X = 37, p = 5$
- $X = 49, p = 7$
- $X = 61, p = 9$
- $X = 72, p = 11$

Q6.

How many lines of symmetry does this rhombus have?



Q7.

Choose the properties that **all rhombuses have** but **parallelograms need not have**.
Select **all** that apply.

All sides are equal.

The opposite sides are parallel.

The opposite angles are equal.

The diagonals bisect each other at right angles.

Q8.

A box **cannot** contain more than 25 apples.

If there are 360 apples, what is the **least** number of such boxes required?

13

14

15

16

Q9.

Peter had 10 kg of potatoes.

He used 20 bags and packed $\frac{1}{4}$ kg of potatoes in each.

He used 6 more bags and packed $\frac{1}{2}$ kg of potatoes in each.

What quantity of potatoes is left unpacked?

Q10.

Susan drew a **scaled pictograph** to show the number of books sold in four weeks of a month.

Week 1	
Week 2	
Week 3	
Week 4	

The number of books sold on Week 4 is **120** more than the number of books sold on Week 2.

What is the scale of the pictograph?

 is 60 books

 is 40 books

 is 30 books

 is 20 books

Q11.

Find the principal invested, if the compound interest at the rate of 10% per annum for 2 years compounded annually is \$1260.

Q12.

Which of the following expressions has a value of 5?

$$5 + 40 \div 5 - 4$$

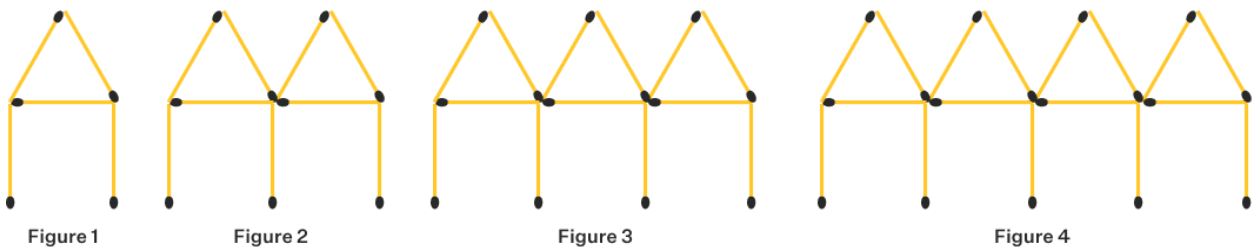
$$- 4 \times 5 + 50 \div 2$$

$$40 - 30 \div 10 + 4$$

$$10 - 6 \div 2 + 3$$

Q13.

Observe the following pattern.



Which figure will have 601 matchsticks?

Figure 600

Figure 300

Figure 150

Figure 125

Q14.

If $z = 4x^2 + 8y^2$, where x and y are odd natural numbers, which of these will **definitely be** a factor of z ? Select **all** that apply.)

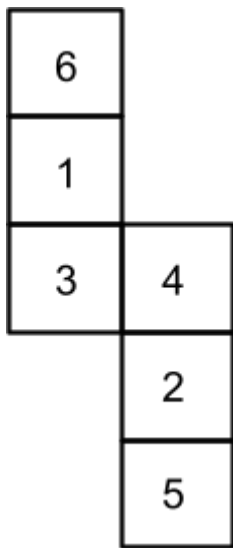
- 2
- 4
- 8
- 16

Q15.

Two natural numbers are in the ratio 3:5. If the difference between their squares is 256, find the greater number.

Q16.

Which number would be on the opposite face of the number 6 if the given net of the cube is folded?



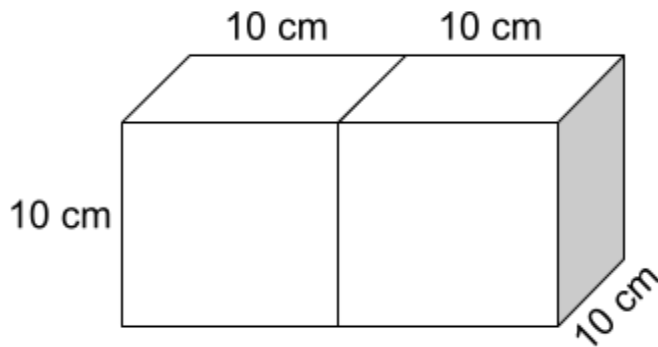
- 4
- 5
- 3
- 2

Q17.

N is a non-zero perfect square such that $N = 15 \times 27 \times A$. What would be the **minimum** value of A?

Q18.

Two cubes each with a side length of 10 cm are joined together as shown.



What will be the surface area of the new shape that is formed?

1200 cm²

1100 cm²

1000 cm²

800 cm²

Q19.

The price of an item is \$50. Its price decreases by 30%.

What is the new price?

\$15

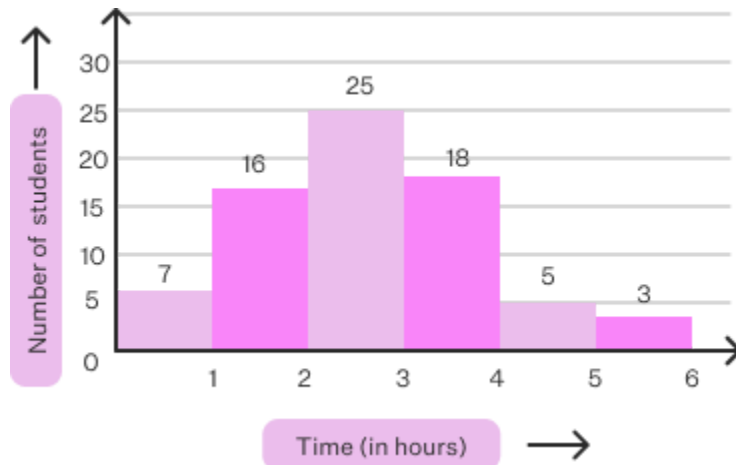
\$20

\$35

\$65

Q20.

The given graph shows the study time of some students.



How many students studied for **less than** 3 hours?

- 23
- 25
- 48
- 66

Answer Key	
1.	5
2.	$\frac{3}{14}$
3.	It will always be less than both p and q.
4.	126 cm
5.	X = 61, p = 9
6.	2
7.	All sides are equal. The diagonals bisect each other at right angles.
8.	15
9.	2 kg
10.	40 books
11.	\$6000
12.	$-4 \times 5 + 50 \div 2$
13.	Figure 150
14.	2, 4
15.	20
16.	3
17.	5
18.	1000 cm ²
19.	\$35
20.	48